Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A straight metal wire of length L, cross-sectional area a and resistivity ρ is made into a square frame. A uniform magnetic field B is perpendicular to the plane of the frame and is changing at a constant rate dB/dt. The current induced in the frame is

(A) $\frac{La}{\rho}\frac{dB}{dt}$

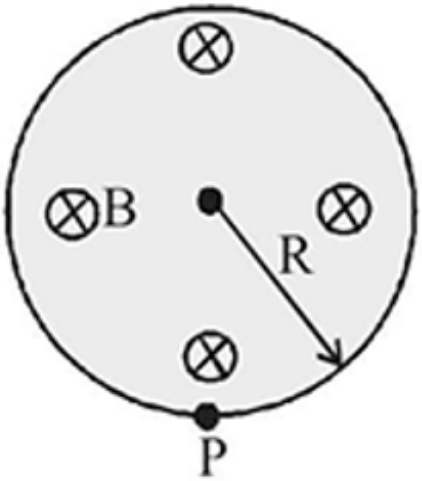
(B) $\frac{La}{4\rho}\frac{dB}{dt}$

(C) $\frac{La}{8\rho}\frac{dB}{dt}$

(D) $\frac{La}{16\rho}\frac{dB}{dt}$

Q.2

A uniform magnetic field of induction B is confined to a cylindrical region of radius R. The magnetic field is increasing at a constant rate of $\frac{\mathrm{dB}}{\mathrm{dt}}$ (tesla/second). An electron of charge e, placed at the point P on the periphery of the field experiences an acceleration.



(A) $\frac{1}{2}\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards left

(B) $\frac{1}{2}\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards right

(C) $\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards left

(D) zero

Q.3

If the flux of magnetic induction through a coil of resistance R and having n turns changes from ϕ_1 to ϕ_2, then the magnitude of the charge that passes through the coil is

(A) $\frac{(\phi_2-\phi_1)}{R}$

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(B) $\frac{n(\phi_2-\phi_1)}{R}$

(C) $\frac{(\phi_2-\phi_1)}{nk}$

(D) $\frac{nR}{(\phi_2-\phi_1)}$

Q.4

A coil of 20×20 cm having 30 turns is making 30 rps in a magnetie field of 1 tesla. The peak value of the induced e.m.f. is approximately

(A) 452 V

(B) 226 V

(C) $113V$

(D) 339 V

Q.5

A short-circuited coil is placed in a time-varying magnetic field. Electrical power is dissipated due to the current induced in the coil. If the number of turns were to be quadrupled and the wire radius halved, the electrical power dissipated would be

(A) cannot be determined

(B) the same

(C) doubled

(D) quadrupled

Q.6

In a uniform magnetic field of induction B a wire in the form of a semicircle of radius r rotates about the diameter of the circle with an angular frequency ω. The axis of rotation is perpendicular to the field. If the total resistance of the circuit is R, the mean power generated per period of rotation is

(A) $\frac{(B\pi r\omega)^2}{2R}$

(B) $\frac{\left(B\pi r^2\omega\right)^2}{8R}$

(C) $\frac{B\pi r^2\omega}{2R}$

(D) $\frac{\left(B\pi r\omega^2\right)^2}{8R}$

Q.7

A metal ring is held horizontally and bar magnet is dropped through the ring with its length along the axis of the ring. The acceleration of the falling magnet

(A) is equal to g

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(B) is less than g

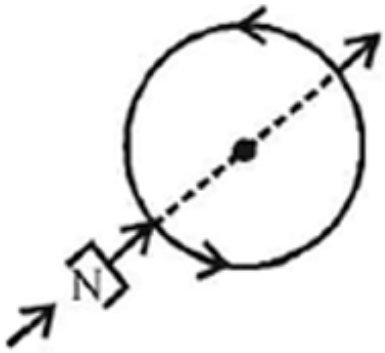
(C) is more than g

(D) depends on the diameter of ring and length of magnet

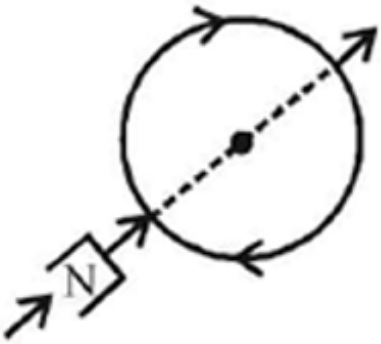
Q.8

Which of the following figure correctly depicts the Lenz's law. The arrows show the movement of the labelled pole of a bar magnet into a closed circular loop and the arrows on the circle show the direction of the induced current

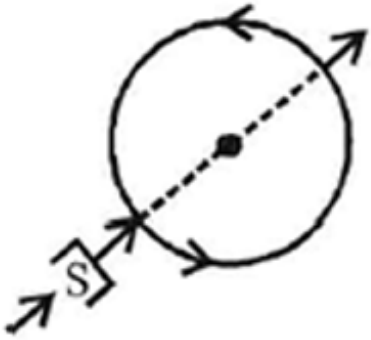
(A)



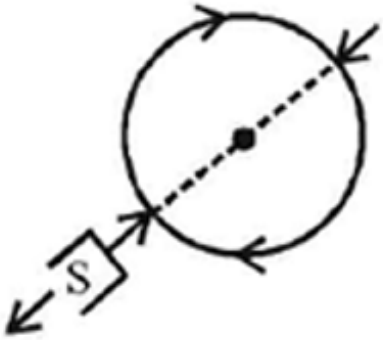
(B)



(C)



(D)

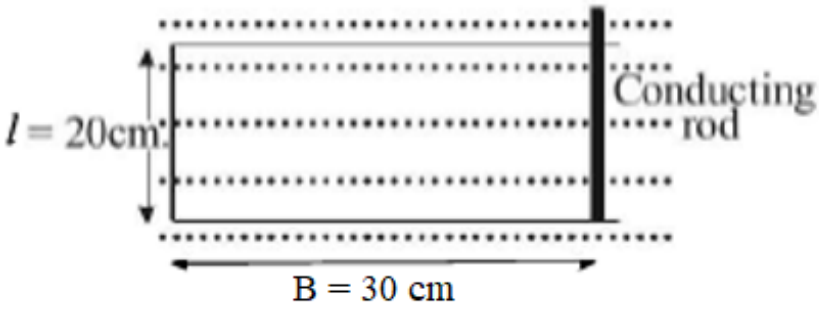


Q.9

Figure shows a conducting rod of negligible resistance that can slide on smooth U-shaped rail made of wire of resistance $1\Omega/\mathrm{m}$. Position of the conducting rod at $\mathrm{t} = 0$ is shown. A time t dependent magnetic field $B = 2t$

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tesla is switched on at $t = 0$



At $t = 0$, when the magnetic field is switched on, the conducting rod is moved to the left at constant speed $5\ \text{cm/s}$ by some external means. The rod moves perpendicular to the rails. At $t = 2\ \text{s}$, induced emf has magnitude

(A) $0.12\ \text{V}$

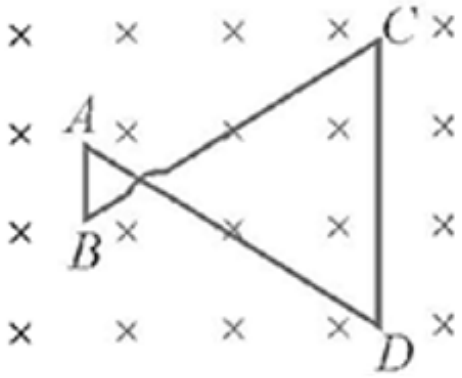
(B) $0.08\ \text{V}$

(C) $0.04\ \text{V}$

(D) $0.02\ \text{V}$

Q.10

A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced current in wires AB and CD are



(A) B to A and D to C

(B) A to B and C to D

(C) A to B and D to C

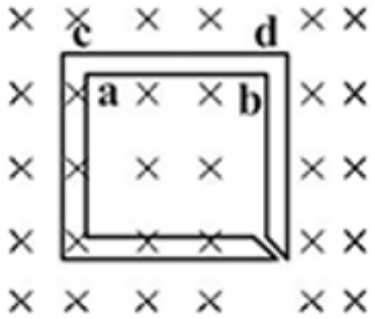
(D) B to A and C to D

Q.11

The figure shows certain wire segments joined together to form a coplanar loop. The loop is placed in a perpendicular magnetic field in the direction going into the plane of the figure.

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The magnitude of the field increases with time. I_1 and I_2 are the currents in the segments ab and cd. Then,



(A) $I_1 > I_2$

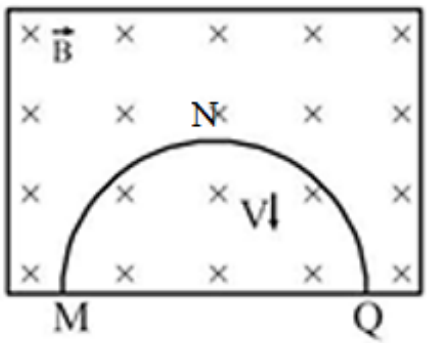
(B) $I_1 < I_2$

(C) I_1 is in the direction ba and I_2 is in the direction cd

(D) I_1 is in the direction ab and I_2 is in the direction dc

Q.12

A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction B. At the position MNQ, the speed of the ring is V and the potential difference developed across the ring is



(A) Zero

(B) $Bv\pi R^2/2$ and M is at higher potential

(C) πRBV and Q is at higher potential

(D) $2RBV$ and Q is at higher potential

Q.13

The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in a time Δt. Then the total quantity of electric charge Q that passes any point in the circuit during the time Δt is represented by

(A) $Q = R \cdot \frac{\Delta\phi}{\Delta t}$

(B) $Q = \frac{1}{R} \cdot \frac{\Delta\phi}{\Delta t}$

(C) $Q = \frac{\Delta\phi}{R}$

(D) $Q = \frac{\Delta\phi}{\Delta t}$

Q.14

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As a result of change in the magnetic flux linked to the closed loop shown in the fig, an e.m.f. V volt is induced in the loop. The work done (joule) in taking a charge Q coulomb once along the loop is



(A) QV

(B) $2QV$

(C) $\mathrm{QV}/2$

(D) Zero

Q.15

A thin circular ring of area A is held perpendicular to a uniform magnetic field of induction B. A small cut is made in the ring and a galvanometer is connected across the ends such that the total resistance of the circuit is R. When the ring is suddenly squeezed to zero area, the charge flowing through the galvanometer is

(A) $\frac{\mathrm{BR}}{\mathrm{A}}$

(B) $\frac{\mathrm{AB}}{\mathrm{R}}$

(C) ABR

(D) $\frac{\mathrm{B}^2\,\mathrm{A}}{\mathrm{R}^2}$

Q.16

A coil having n turns and resistance $\mathrm{R}\Omega$ is connected with a galvanometer of resistance $4\mathrm{R}\Omega$. This combination is moved in time t seconds from a magnetic field W_1 weber to W_2 weber. The induced current in the circuit is

(A) $-\frac{(\mathrm{W}_1-\mathrm{W}_2)}{\mathrm{Rnt}}$

(B) $-\frac{\mathrm{n}(\mathrm{W}_2-\mathrm{W}_1)}{5\mathrm{Rt}}$

(C) $-\frac{(\mathrm{W}_2-\mathrm{W}_1)}{5\mathrm{Rnt}}$

(D) $-\frac{\mathrm{n}(\mathrm{W}_2-\mathrm{W}_1)}{\mathrm{Rt}}$

Q.17

The magnetic flux (in weber) linked with a coil of resistance 10Ω is varying with respect to time t as $\phi = 4t^2 + 2t + 1$. Find the current (in A) in the coil at time $\mathrm{t} = 1$ second.

(A) 1

(B) 2

(C) 3

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(D) 4

Q.18

A conducting square loop is placed in a magnetic field B with its plane perpendicular to the field. The sides of the loop start shrinking at a constant rate α. The induced emf in the loop at an instant when its side is ' a ' is

(A) $2a\alpha B$

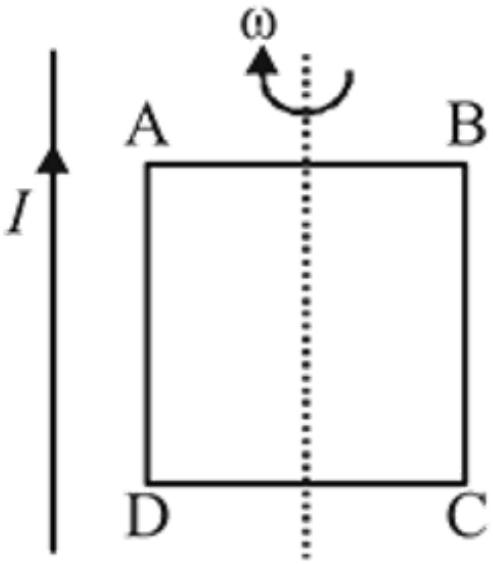
(B) $a^2\alpha B$

(C) $2a^2\alpha B$

(D) $a\alpha B$

Q.19

A rectangular loop is present in the magnetic field region of an infinite long wire. Now the loop is being rotated as shown in the figure. Then the induced current in side AD will be



(A) along DA

(B) along AD

(C) zero

(D) None of these

Q.20

If the current through an inductor of 2 H is given by $I = t\sin t$ A, then the voltage across the inductor is

(A) $\cos t + t\sin t$

(B) $2t\cos t + 2\sin t$

(C) $t\cos t + \sin t$

(D) $2t\sin t + 2\cos t$

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Q.1 Answer:

$\frac{La}{16\rho}\frac{dB}{dt}$

Solution:

Side of frame $= \frac{L}{4}$**. Area of frame is** $A = L^2/16$**. Magnitude of induced emf is**

$|e| = \frac{d\phi}{dt} = \frac{d}{dt}(AB) = A\frac{dB}{dt} = \frac{L^2}{16}\frac{dB}{dt}$

Resistance of frame is $R = \frac{\rho L}{a}$**.**

$\therefore$ **Induced current** $= \frac{|e|}{R} = \frac{La}{16\rho}\frac{dB}{dt}$

Q.2 Answer:

$\frac{1}{2}\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards left

Solution:

If we consider the cylindrical surface to be a ring of radius R**, there will be an induced emf due to changing field.**

$$\int \mathrm{E}\mathrm{d}\vec{\ell} = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = -\mathrm{A}\frac{\mathrm{dB}}{\mathrm{dt}}$$

$\Rightarrow \mathrm{E}(2\pi\mathrm{R}) = -\mathrm{A}\frac{\mathrm{dB}}{\mathrm{dt}} = -\pi\mathrm{R}^2\frac{\mathrm{dB}}{\mathrm{dt}} \Rightarrow \mathrm{E} = -\frac{\mathrm{R}}{2}\frac{\mathrm{dB}}{\mathrm{dt}}$

$\therefore$ **Force on the electron**

$\mathrm{F} = -\mathrm{Ee} = \frac{\mathrm{eR}}{2}\frac{\mathrm{dB}}{\mathrm{dt}}$

$\Rightarrow$ **acceleration** $= \frac{1}{2}\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$

As the field is increasing being directed inside the paper, hence there will be anticlockwise induced current (in order to oppose the cause) in the ring (assumed). Hence there will be a force towards left on the electrons.

Q.3 Answer:

$\frac{n(\phi_2-\phi_1)}{R}$

Solution:

Induced is emf is $|e| = n\frac{\Delta\phi}{\Delta t}$**. Now** $\Delta q = I\Delta t = \frac{e}{R}\Delta t = \frac{n\Delta\phi}{R\Delta t} \times \Delta t = \frac{n\Delta\phi}{R} = \frac{n(\phi_2-\phi_1)}{R}$ $\therefore$ **(b)**

Q.4 Answer:

226 V

Solution:

$e_0 = NAD\mathrm{o}$

$= 30 \times 400 \times 10^{-4} \times 1 \times 30 \times 2\pi = 226\mathrm{volt}$

$\therefore$ **(b)**

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Q.5 Answer:

the same

Solution:

$P = \frac{E^2}{R} = \frac{\pi r^2}{\rho \ell}\left(\frac{d\phi}{dt}\right)^2 = \frac{\pi r^2}{\rho \ell}\left[\frac{d}{dt}(NBA)^2\right]$

$= \frac{\pi r^2}{\rho \ell} N^2 A^2 \left(\frac{dB}{dt}\right)^2 \Rightarrow P \propto \frac{N^2 r^2}{\ell}$

Case 1: $P_1 \propto \frac{N^2 r^2}{\ell}$

Case 2: $P_2 \propto \frac{(4N)^2 (r/2)^2}{4\ell}$

Note : When we decrease the radius of the wire, its length increases but volume remains the same

$\Rightarrow \frac{P_1}{P_2} = \frac{1}{1}$ **$\therefore$ Power remains the same.**

Q.6 Answer:

$\frac{\left(B\pi r^2 \omega\right)^2}{8R}$

Solution:

$\phi = \vec{B} \cdot \vec{A}; \phi = BA\cos\omega t$

$\varepsilon = -\frac{d\phi}{dt} = \omega BA \sin\omega t; \quad i = \frac{\omega BA}{R}\sin\omega t$

$P_{\text{inst}} = i^2 R = \left(\frac{\omega BA}{R}\right)^2 \times R\sin^2\omega t$

$P_{\text{avg}} = \frac{\int_0^T P_{\text{inst}} \times dt}{\int_0^T dt} = \frac{(\omega BA)^2}{R}\frac{\int_0^T \sin^2\omega t dt}{\int_0^T dt} = \frac{1}{2}\frac{(\omega BA)^2}{R}$

$\therefore P_{\text{avg}} = \frac{\left(\omega B \pi r^2\right)^2}{8R} \quad \left[A = \frac{\pi r^2}{2}\right]$

Q.7 Answer:

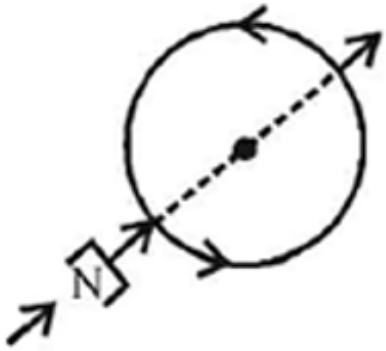
is less than g

Solution:

A metal ring is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring. The magnet falls with an acceleration less than g

As the magnet falls, the magnetic flux linked with the ring increases. This induces emf in the ring which opposes the motion of the falling magnet, hence $\mathrm{a} < \mathrm{g}$.

Q.8 Answer:



Solution:

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When a north pole of a bar magnet moves towards the coil, the induced current in the coil flows in a direction such that the coil presents its north pole to the bar magnet as shown in figure (a). Therefore, the induced current flows in the coil in the anticlockwise direction. When a north pole of a bar magnet moves away from the coil, the induced current in the coil flows in a direction such that the coil presents its south pole to the bar magnet as shown in figure (b).



Therefore induced current flows in the coil in the clockwise direction.

Q.9 Answer:

$0.08\ \mathrm{V}$

Solution:

At $\mathrm{t}=2\ \mathrm{s},\ \mathrm{B}=4\ \mathrm{T}, \frac{\mathrm{dB}}{\mathrm{dt}}=\frac{2\ \mathrm{T}}{\mathrm{s}}$

$\mathrm{A}=20\times 30\ \mathrm{cm}^2$

$=600\times 10^{-4}\ \mathrm{m}^2, \frac{\mathrm{dA}}{\mathrm{dt}}=-(5\times 20)\mathrm{cm}^2/\mathrm{s}$

$=-100\times 10^{-4}\ \mathrm{m}^2/\mathrm{s}$

$\mathrm{E}=-\frac{\mathrm{d}\phi}{\mathrm{dt}}=-\left[\frac{\mathrm{d(BA)}}{\mathrm{dt}}\right]=-\left[\frac{\mathrm{BdA}}{\mathrm{dt}}+\frac{\mathrm{AdB}}{\mathrm{dt}}\right]$

$=-\left[4\times\left(-100\times 10^{-4}\right)+600\times 10^{-4}\times 2\right]$

$=-[-0.04+0.120]=-0.08\ \mathrm{V}$

Q.10 Answer:

B to A and D to C

Solution:

As the magnetic field increases, its flux also increases into the page and so induced current in bigger loop will be anticlockwise. i.e., from D **to** C **in bigger loop and then from** B **to** A **in smaller loop.**

Q.11 Answer:

I_1 is in the direction ab and I_2 is in the direction dc

Solution:

The magnetic field is increasing in the downward direction. Therefore, according to Lenz's law the current I_1 **will flow in the direction** ab **and** I_2 **in the direction dc.**

Q.12 Answer:

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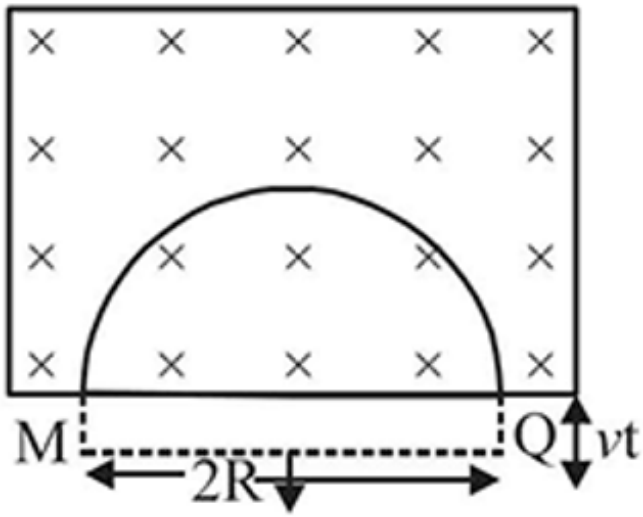
$2RBV$ and Q is at higher potential

Solution:

Rate of decrease of area of the semicircular ring

$-\frac{dA}{dt} = (2R)V$

Acoording to Faraday's law of induction induced emf $e = -\frac{d\phi}{dt} = -B\frac{dA}{dt} = -B(2RV)$



The induced current in the ring must generate magneticfield in the upward direction. Thus Q is at higher potential.

Q.13 Answer:

$Q = \frac{\Delta\phi}{R}$

Solution:

$\frac{\Delta\phi}{\Delta t} = \varepsilon = iR \Rightarrow \Delta\phi = (i\Delta t)R = QR$

$\Rightarrow Q = \frac{\Delta\phi}{R}$

Q.14 Answer:

QV

Solution:

$\xi = \frac{W}{Q} \Rightarrow V = \frac{W}{Q} \Rightarrow W = QV$

Q.15 Answer:

$\frac{AB}{R}$

Solution:

The individual emf produced in the coil $e = \frac{-d\phi}{dt}$

$\therefore$ The current induced will be $i = \frac{|e|}{R} \Rightarrow i = \frac{1}{R}\frac{d\phi}{dt}$

But $i = \frac{dq}{dt} \Rightarrow \frac{dq}{dt} = \frac{1}{R}\frac{d\phi}{dt} \Rightarrow \int dq = \frac{1}{R}\int d\phi \Rightarrow q = \frac{BA}{R}$

Q.16 Answer:

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$-\frac{n(W_2-W_1)}{5Rt}$

Solution:

$\frac{d\phi}{dt}=\frac{(W_2-W_1)}{t}R_{tot}=(R+4R)\Omega=5R\Omega$

$i=\frac{nd\varphi}{R_{tot}dt}=\frac{-n(W_2-W_1)}{5Rt}$

$(\because W_2 \& W_1$ **are magnetic flux)**

Q.17 Answer:

1

Solution:

Given : $\phi=4t^2+2t+1$ **wb**

$\therefore \quad \frac{d\phi}{dt}=\frac{d}{dt}\left(4t^2+2t+1\right)=8t+2=|\varepsilon|$

Induced current, $I=\frac{|\varepsilon|}{R}=\frac{8t+2}{10\Omega}=\frac{8t+2}{10}$ A

At $t=1$ s

$I=\frac{8\times1+2}{10}$ A $=1$ A

Q.18 Answer:

$2a\alpha B$

Solution:

At any time t**, the side of the square** $a=(a_0-\alpha t)$**, where** $a_0=$ **side at** $t=0$**. At this instant, flux through the square :**

$\phi=BA\cos0^\circ=B\left(a_0-\alpha t\right)^2$

$\therefore$ **emf induced** $E=-\frac{d\phi}{dt}$

$\Rightarrow E=-B.2\left(a_0-\alpha t\right)(0-\alpha)=+2\alpha aB$

Q.19 Answer:

along DA

Solution:

Apply Lenz's law such that the net force on loop will be towards the infinite long wire.

Q.20 Answer:

$2t\cos t+2\sin t$

Solution:

$e_L=L\frac{di}{dt}$

$=L t\cos t+\sin t \qquad L=2$ H

$=2t\cos t+2\sin t$

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